

# Chapter 157 — Diagnosis and Staging of Prostate Cancer

- **Early prostate cancer is rarely symptomatic**; symptoms imply locally advanced or metastatic disease.
- **Locally advanced disease**: obstructive urinary symptoms, ureteric obstruction (pain or renal failure), haematuria, reduced ejaculate volume and, rarely, erectile dysfunction.
- **Metastatic disease**: bone pain, pathological fracture, lower-limb oedema; less commonly malignant retroperitoneal fibrosis, paraneoplastic syndromes, disseminated intravascular coagulation (DIC) and paralysis from spinal cord compression.
- **Advanced presentation is now uncommon** owing to screening with prostate-specific antigen (PSA) and digital rectal examination (DRE); diagnosis is usually by **transrectal ultrasound (TRUS)-guided needle biopsy before symptoms**.
- **Staging defines extent and risk** for management and prognosis: PSA, DRE, biopsy **grade and cancer volume**, and imaging where indicated.

## DIAGNOSIS

### Screening

#### General Concepts of Screening

- **Screening** tests healthy, asymptomatic populations; **diagnosis** identifies disease in people with signs or symptoms.
- **Early diagnosis is not invariably beneficial**: screening harms through **overdiagnosis and overtreatment**.
- Screening **reduces advanced disease at diagnosis** and, in recent data, **prostate cancer mortality**; the dispute is the **balance of benefit and harm**, which depends on the individual's characteristics and preferences.
- Since PSA testing spread (early 1990s), **prostate cancer mortality has fallen by more than 40%** and advanced disease at diagnosis by about three-quarters (Surveillance, Epidemiology, and End Results [SEER]); modelling attributes **half to two-thirds of the mortality fall** to screening.
- **Randomised trials of screened versus unscreened populations** are the highest level of evidence.

#### Randomised Trials

##### COMPARISON — THE TWO SCREENING TRIALS: ERSPC VS PLCO

|                            | ERSPC                                                       | PLCO                                                                                           |
|----------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Name                       | European Randomised Study of Screening for Prostate Cancer  | Prostate, Lung, Colorectal and Ovarian Cancer Screening Trial (National Cancer Institute, NCI) |
| Started · primary endpoint | 1993 · prostate cancer-specific mortality                   | 1993 · prostate cancer-specific mortality                                                      |
| Median follow-up           | 16 years                                                    | 17 years                                                                                       |
| Mortality result           | <b>20% relative reduction</b>                               | <b>No difference</b>                                                                           |
| Other effects              | Fewer high-grade and locally advanced or metastatic cancers | —                                                                                              |
| To prevent one death       | <b>Invite 570, diagnose 18 extra</b> (16 years)             | —                                                                                              |

| ERSPC          | PLCO                                                                                                |
|----------------|-----------------------------------------------------------------------------------------------------|
| Why it differs | —                                                                                                   |
|                | <b>Contamination</b> (control-arm PSA testing) and <b>threefold fewer events</b> → <b>low power</b> |

- Both trials show **overdiagnosis** (cancers otherwise never detected) and overtreatment — **worst after 65 years**, where a randomised trial of radical prostatectomy versus watchful waiting showed minimal benefit; **average age at diagnosis is about 66**.

**Concept** — *The ERSPC–PLCO discordance is a matter of trial design, not biology: control-arm contamination and too few events can erase a real screening effect.*

## Specialty Group Recommendations

### COMPARISON — SCREENING RECOMMENDATIONS BY BODY

| Body                                                       | Who and when                                                                                                                                               | Stance                                                                                    |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| U.S. Preventive Services Task Force (USPSTF) 2008          | <75 years · ≥75 years                                                                                                                                      | Insufficient evidence · <b>against</b>                                                    |
| USPSTF 2012 (draft 2011)                                   | All ages                                                                                                                                                   | <b>Grade D — against screening</b>                                                        |
| USPSTF 2018 (draft 2017)                                   | <b>55–69 years</b> · ≥70 years                                                                                                                             | <b>Grade C — shared decision making</b> · against                                         |
| American College of Preventive Medicine                    | Population                                                                                                                                                 | Insufficient evidence for routine PSA + DRE                                               |
| American Cancer Society                                    | Life expectancy ≥10 years; <b>from 50</b> (average risk); <b>earlier</b> for Black people or a first-degree relative (father, brother) diagnosed <65 years | Informed decision                                                                         |
| American Urological Association (AUA) 2013, validated 2018 | <b>55–69 years</b>                                                                                                                                         | Shared decision making; <b>2-yearly interval</b> (from ERSPC) keeps benefit, reduces harm |
| National Comprehensive Cancer Network (NCCN)               | <b>From 45 years</b>                                                                                                                                       | Risk–benefit discussion, baseline PSA; repeat every <b>1–4 years by PSA result</b>        |

- USPSTF credits **active surveillance** (observation with selective delayed treatment) with reducing screening harms.
- Ages and interval **will stay debated**: further randomised trials are logistically unlikely.
- **Opportunistic screening is prevalent** in the United States; adults at periodic health examinations should be told the benefits and risks.

**Concept** — *Every body converges on informed or shared decision making; they differ only on the age window — NCCN starts earliest (45), USPSTF stops most clearly (70).*

## Diagnostic Modalities

### Digital Rectal Examination

- Before PSA, **DRE was the only detection tool**.
- **Limitations**: fair reproducibility even among experts; misses many early cancers; **little value at PSA <3.0 ng/mL**; no proven mortality benefit, and some advise against routine screening DRE.
- **Drawbacks**: separate attendance (blood allows reflex testing on one sample), cost, and it **deters screening uptake**.
- **Why it persists**: an abnormal DRE carries real cancer risk and the examination is simple, so most urologists **combine PSA and DRE**; many risk nomograms include DRE.

## Prostate-Specific Antigen

- **Human kallikrein-family protease** secreted at very high concentration (mg/mL) into seminal fluid; circulates **complexed (bound) and free**, measured by U.S. Food and Drug Administration (FDA)-approved assays.

### Factors influencing PSA

#### COMPARISON — WHAT MOVES SERUM PSA

| Factor                                                                                                                    | Effect           | Point to hold                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Prostate disease</b> — cancer, benign prostatic hyperplasia (BPH), prostatitis                                         | ↑                | <b>Most important factor</b> ; not cancer-specific; disease possible with normal PSA                             |
| Prostate volume · age                                                                                                     | ↑ · ↑            | Volume explains far more PSA variance than age                                                                   |
| Black ethnicity                                                                                                           | ↑                | Higher PSA even without cancer                                                                                   |
| Massage, biopsy, transurethral resection                                                                                  | ↑                | Architectural disruption releases PSA into blood                                                                 |
| DRE                                                                                                                       | Slight ↑         | Within assay error; <b>rarely a false positive</b>                                                               |
| Ejaculation                                                                                                               | Conflicting      | If minimally raised, <b>repeat after 48 hours' abstinence</b>                                                    |
| Hypogonadism                                                                                                              | ↓                | <b>Androgen-dependent</b> (appears at puberty with rising luteinising hormone and testosterone); may mask cancer |
| Obesity · statins                                                                                                         | Slight ↓ · may ↓ | Obesity by haemodilution                                                                                         |
| 5α-reductase inhibitors                                                                                                   | ↓                | See below                                                                                                        |
| BPH surgery                                                                                                               | ↓                | Removes the transition zone; <b>resets baseline</b> variably                                                     |
| Cancer treatment — luteinising hormone-releasing hormone agonists, orchidectomy, radiotherapy, prostatectomy, cryotherapy | ↓                | Less epithelium, less PSA per cell                                                                               |
| Change of assay                                                                                                           | Variable         | <b>Pseudo-acceleration or pseudo-deceleration</b> on serial testing                                              |
| Genetics                                                                                                                  | Variable         | Genetically adjusted PSA proposed                                                                                |

- **5α-reductase inhibitors lower PSA**: finasteride (type 2) and dutasteride (types 1 and 2); **finasteride 1 mg for alopecia lowers PSA as much as 5 mg for BPH**.
- **Baseline PSA before starting**, then serial measurement.
- Correction: **×2 after ≥12 months**; response varies, so **×2.3 at 2 years, ×2.5 at 7 years** proposed; or take the **on-treatment nadir as new baseline** and investigate any rise.
- **Interpret every PSA against** age, urinary tract infection or prostate disease, recent procedures and prostate-directed treatment.

### Clinical use for diagnosis

- Original FDA-approved assays (1994) measured **free PSA (fPSA) plus PSA complexed to α1-antichymotrypsin (cPSA)** = total PSA (tPSA); separate fPSA and cPSA assays followed.
- PSA testing detects **more cancers, and more organ-confined cancers**, than detection without it.
- **Risk is continuous**: future cancer risk and the present chance of a positive biopsy both rise incrementally with PSA across its full range (Table 157.1); baseline PSA predicts later detection.

- **Malmö Preventive Project: PSA measured in the 40s predicts metastasis and death more than 25 years later.**

**TABLE 157.1 PROSTATE CANCER DETECTION AS A FUNCTION OF SERUM PROSTATE-SPECIFIC ANTIGEN LEVEL AND DIGITAL RECTAL EXAMINATION FINDINGS IN CONTEMPORARY SERIES**

| PSA (ng/mL) | Yield on biopsy, DRE non-suspicious (%) | Yield on biopsy, DRE suspicious (%) | High-grade (Gleason $\geq 7$ ) on biopsy (%) |
|-------------|-----------------------------------------|-------------------------------------|----------------------------------------------|
| 0–1         | 8.8                                     | —                                   | 0.9                                          |
| 1–2         | 17.0                                    | —                                   | 2.0                                          |
| 0–2         | 8–12                                    | —                                   | 1.4                                          |
| 2–4         | 15–25                                   | —                                   | 5.2                                          |
| <4          | 15                                      | 13–17                               | 2.3 (DRE –)                                  |
| 4–10        | 17–32                                   | <b>45–51</b>                        | 4.1 (–) · 11.7 (+)                           |
| >4          | 23–38                                   | 55–63                               | 5.8 (–) · 20.6 (+)                           |
| >10         | 43–65                                   | <b>70–90</b>                        | 19.4 (–) · <b>50.5 (+)</b>                   |

Data from several contemporary series (Andriole et al., 2005; Catalona et al., 1998; Crawford et al., 1996; Schröder et al., 1998; Thompson et al., 2004). Yield = cancers ÷ men biopsied; detection rate (cancers ÷ men screened) is far lower: 1–3% at PSA <4, 11–27% at 4–10, 31–76% at >10.

- **What it is for:** PSA and DRE are **independent, additive** indicators, and **no PSA level carries zero risk**.
- **Pattern:** yield and high-grade rate climb with every PSA band; a **suspicious DRE roughly doubles the yield** at a given PSA and multiplies the high-grade rate.
- **Traps:** PSA 0–1 ng/mL still yields ~9% (Prostate Cancer Prevention Trial, PCPT) — **"normal" PSA does not exclude cancer; detection rate is not yield**.

#### *Triggers for prostate biopsy*

- **No agreed threshold:** dichotomising PSA discards information; **PCPT shows risk is continuous**, so some avoid calling PSA "elevated" or "abnormal".
- **Trade-off:** a higher threshold misses curable cancer; a lower threshold adds unnecessary biopsies and overdiagnosis.
- **Thresholds:** 4 ng/mL original FDA upper limit; many clinicians use **2.5–3 ng/mL**; most ERSPC centres used 3 ng/mL; 10 ng/mL suggested **over 70 years** to reduce harm (AUA).
- **Repeat a suspicious PSA before biopsy** — physiological fluctuation causes false positives.
- **No empirical antibiotics for a raised PSA alone** without urinary symptoms (Choosing Wisely).
- Combine PSA with **family history, ethnicity and DRE**.

#### **Prostate-Specific Antigen Derivatives and Molecular Forms**

- All aim to raise **specificity**, by **volume adjustment, kinetics or molecular form**.
- **Volume** (ultrasonography or magnetic resonance imaging [MRI]) corrects for common BPH-driven PSA rises.
- **Why fPSA is lower in cancer:** differential isoform expression — **transition zone (origin of BPH) vs peripheral zone (origin of most cancers)** — so cancer carries more complexed PSA and a **lower percentage free (%fPSA)**.
- **Isoform sequence:** proPSA (precursor, secreted by luminal epithelium) → active fPSA → cleaved to **benign PSA (bPSA) or intact PSA (iPSA)**, both inactive and uncomplexed.

**COMPARISON — PSA DERIVATIVES AND MOLECULAR FORMS**

| Marker                                   | Definition                                                    | Cut-off / finding                                                                                                                                                                               | Use and limits                                                                                                                                                             |
|------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PSA density (PSAD)</b>                | PSA ÷ prostate volume                                         | <b>≥0.15 → biopsy</b> (PSA 4–10, normal DRE)                                                                                                                                                    | Counselling at 4–10; repeat biopsy; <b>active surveillance eligibility</b> ; inconsistently validated                                                                      |
| Complexed PSAD · transition zone density | cPSA ÷ volume · PSA ÷ transition zone volume                  | —                                                                                                                                                                                               | Transition zone is the main PSA source without cancer                                                                                                                      |
| <b>PSA velocity (PSAV)</b>               | PSA change over time                                          | <b>&gt;0.75 ng/mL/yr</b> (PSA 4–10); <b>0.4</b> at lower PSA                                                                                                                                    | Confounded by physiological fluctuation, calculation method and timing                                                                                                     |
| PSAV — prognostic                        | —                                                             | <b>&gt;0.35 ng/mL/yr</b> 10–15 years before diagnosis → <b>life-threatening cancer</b> ; <b>&gt;2 ng/mL/yr</b> in the year before diagnosis → <b>cancer death after surgery or radiotherapy</b> | Systematic review: no gain over PSA alone; later studies: some gain — unresolved                                                                                           |
| <b>%fPSA</b>                             | fPSA ÷ tPSA; low in cancer                                    | <b>25%</b> (PSA 4–10, benign DRE): ~95% of cancers found, ~20% of biopsies spared                                                                                                               | Best at 4–10, unclear <4; <b>no volume needed</b> ; <b>unaffected by ethnicity or 5α-reductase inhibitors</b> ; rises with age and volume; assay- and gland-size-dependent |
| <b>cPSA</b>                              | Complexed fraction; high in cancer                            | —                                                                                                                                                                                               | At high sensitivity: <b>specificity &gt; tPSA, ≈ %fPSA</b> (PSA 4–10 and 2.6–4.0)                                                                                          |
| proPSA                                   | Precursor; relatively high in cancer                          | —                                                                                                                                                                                               | May help at PSA 2–10; not always additive to %fPSA                                                                                                                         |
| bPSA                                     | Found in nodular BPH tissue                                   | —                                                                                                                                                                                               | <b>Marker of BPH</b>                                                                                                                                                       |
| <b>Human kallikrein 2 (hK2)</b>          | Related serine protease; high in poorly differentiated cancer | —                                                                                                                                                                                               | Inconsistent for detection; <b>tracks grade and volume</b>                                                                                                                 |

- At 95% sensitivity, **%fPSA 25% matches PSAD 0.078 in specificity** without a volume measurement.

**Multiplex tests**

- **Prostate Health Index (phi)**, FDA-approved: tPSA + fPSA + [–2]proPSA; **outperforms tPSA and %fPSA** for any and high-grade cancer in prospective trials.
- **4Kscore**: tPSA, fPSA, iPSA and hK2; **predicts clinically significant cancer**.

**Other Markers****COMPARISON — BLOOD, URINE AND TISSUE MARKERS BEYOND PSA**

| Test      | Sample     | Basis                                                                    | Bottom line                                                     |
|-----------|------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
| IsoPSA    | Serum      | Cancer-specific structural PSA isoforms (aqueous two-phase partitioning) | Validated ≥50 years, tPSA ≥4 ng/mL; <b>beats tPSA and %fPSA</b> |
| SelectMDx | Urine mRNA | HOXC6, DLX1                                                              | Biopsy-naïve: spares biopsies, few high-grade missed            |
| ExoDx     |            | Three-gene expression                                                    |                                                                 |

| Test                                 | Sample                      | Basis                                                    | Bottom line                                                                               |
|--------------------------------------|-----------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                                      | Urine<br>exosome<br>RNA     |                                                          | Separates <b>Gleason ≥7</b> from Gleason 6 and benign; avoids about a quarter of biopsies |
| <b>Prostate cancer gene 3 (PCA3)</b> | Urine mRNA                  | Noncoding prostate-specific mRNA overexpressed in cancer | Predicts positive <b>initial or repeat</b> biopsy; <b>conflicting for aggressiveness</b>  |
| TMPRSS2:ERG                          | Urine                       | Gene fusion (cf. BCR-ABL in leukaemia)                   | Higher specificity; <b>conflicting for aggressiveness</b>                                 |
| MyProstateScore (MPS)                | Urine + serum               | tPSA + PCA3 + TMPRSS2:ERG                                | About half spared immediate biopsy                                                        |
| ConfirmMDx                           | <b>Benign biopsy tissue</b> | Multiplex DNA methylation                                | <b>After a negative biopsy:</b> high negative predictive value → avoids repeat biopsy     |

- **tPSA alone rarely triggers biopsy:** add serum or urine biomarkers plus age, family history and ethnicity.

**Concept** — Every refinement attacks one weakness: **PSA is organ-specific, not cancer-specific**. Volume corrects for BPH, kinetics for noise, molecular forms for zone of origin — none replaces biopsy, but together they select who needs one.

## STAGING

### General Concepts of Staging

- **Clinical staging predicts extent from pretreatment parameters** — DRE, PSA and derivatives, biopsy, imaging — for prognosis and treatment choice.

### Clinical Versus Pathological Staging

|       | Clinical stage                                             | Pathological stage                                                                             |
|-------|------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Basis | DRE, PSA, needle biopsy, imaging — <b>before treatment</b> | Histology of prostate, seminal vesicles and pelvic nodes (if dissected) — <b>after removal</b> |
| Value | Guides initial decisions                                   | <b>More accurate; better predicts outcome</b>                                                  |

- **Biochemical recurrence-free and cancer-specific survival both fall as pathological stage rises.**
- **Pathological predictors after radical prostatectomy:** tumour grade, surgical margin status, extracapsular extension, seminal vesicle invasion, pelvic node involvement.

### Classifications

- **Whitmore-Jewett** system: historical interest only.
- **TNM (tumour, node, metastasis)** is current — adopted by the American Joint Committee on Cancer in **1975**, repeatedly revised.
- The latest version builds **prognostic stage groups from TNM + PSA + grade group 1–5**.

TABLE 157.2 2017 TNM CLINICAL STAGING SYSTEMS FOR PROSTATE CANCER

| Category                       | Stage           | Definition                                                                   |
|--------------------------------|-----------------|------------------------------------------------------------------------------|
| <b>cT — not palpable</b>       | TX · T0         | Cannot be assessed · no evidence of primary                                  |
|                                | T1a · T1b       | Incidental at resection: ≤5% · >5% of resected tissue                        |
|                                | T1c             | <b>Found by needle biopsy</b> (one or both sides), not palpable              |
| <b>cT — palpable, confined</b> | T2a · T2b · T2c | ≤ <b>half of one side</b> · > half of one side, not both · <b>both sides</b> |
| <b>cT — extraprostatic</b>     | T3a             | Extraprostatic extension (unilateral or bilateral), not fixed                |

| Category | Stage           | Definition                                                                                                                                  |
|----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|          | T3b             | <b>Seminal vesicle invasion</b>                                                                                                             |
|          | T4              | <b>Fixed</b> , or invades structures <b>other than seminal vesicles</b> : external sphincter, rectum, bladder, levator muscles, pelvic wall |
| cN       | NX · N0 · N1    | Not assessed · no regional nodes · <b>regional node metastasis</b>                                                                          |
| cM       | MX · M0         | Not assessed · none                                                                                                                         |
|          | M1a · M1b · M1c | <b>Non-regional nodes · bone</b> · other sites ± bone                                                                                       |

- **What it is for**: clinical T stage is set by **palpability**, and T1 by how the cancer was found.
- **Pattern**: T1 impalpable → T2 palpable and confined → T3 beyond capsule → T4 fixed or invading neighbours.
- **Traps**: **seminal vesicle invasion is T3b, not T4**; **non-regional nodes are M1a, not N1**; bilateral impalpable biopsy-detected cancer stays **T1c, not T2c**.

## Prediction of Tumour Extent

### Prostate-Specific Antigen

- PSA rises with **pathological stage** (volume link disputed) but **cannot stage an individual**: stages overlap, BPH contributes, and **poorly differentiated tumours make less PSA per gram**.
- **Organ-confined disease**: ~80% at PSA <4, ~66% at 4–10, <50% at >10 ng/mL.
- **Pelvic node involvement**: ~20% at PSA >20, ~75% at PSA >50 ng/mL.
- PSA derivatives and newer PSA-based tests are also studied for grade and extent.
- **Prostatic acid phosphatase** relates to stage and progression but has been **displaced by PSA**.

### Digital Rectal Examination

- Establishes palpability → **clinical T stage**; an abnormal DRE raises the risk of **grade group 4–5** cancer in screened men.
- **Poor sensitivity and reproducibility** → **over- and understaging**; useful only combined with other parameters.

### Prostate Needle Biopsy

- **Histological grade is the most important biopsy finding**; the **Gleason system** is the most widely used.
- **Gleason score**: at low power, grade (1–5) the **predominant pattern (largest area)** → grade the **second most common** → sum = 2–10.
- **Tertiary pattern 5** (2005 consensus): 3 + 4 or 4 + 3 with tertiary 5 is recorded as **3 + 5 or 4 + 5**.
- Scores 2–5 are no longer assigned, so the **lowest score in practice is 6** — confusing to patients and a **barrier to active surveillance** → **grade groups**:

| Grade group   | 1  | 2         | 3         | 4                       | 5    |
|---------------|----|-----------|-----------|-------------------------|------|
| Gleason score | ≤6 | 3 + 4 = 7 | 4 + 3 = 7 | 8 (4 + 4, 3 + 5, 5 + 3) | 9–10 |

- Grade groups simplify grading, **label the lowest risk as group 1 rather than "score 6"**, and predict prognosis accurately.
- **Grade is never used alone**: number and percentage of **positive cores** and **intraductal or cribriform patterns** are prognostic and guide active surveillance selection.

- **Seminal vesicle or periprostatic fat on biopsy → worse prognosis;** seminal vesicle or capsular biopsy: routine for some, **only for a large palpable basal tumour** for others.

**Concept** — *Grade groups renumbered the Gleason scale without re-measuring it: the change is communicative, and the examinable point is that 3 + 4 and 4 + 3 are different groups.*

### Combined Use of Pretreatment Parameters

- **Nomograms combining T stage, PSA and Gleason score or grade group** predict extent and long-term outcome better than any single parameter.

#### COMPARISON — D'AMICO RISK GROUPS

| Risk group   | Criteria                           | Freedom from disease at 10 years after radical prostatectomy |
|--------------|------------------------------------|--------------------------------------------------------------|
| Low          | T1–T2a and PSA ≤10 and Gleason ≤6  | 83%                                                          |
| Intermediate | T2b or PSA >10 to <20 or Gleason 7 | 46%                                                          |
| High         | T2c or PSA >20 or Gleason 8–10     | 29%                                                          |

- **Trap:** low risk needs **all three**; intermediate or high needs **any one**.
- Later validated schemes include the **Cancer of the Prostate Risk Assessment (CAPRA) score**. Multiparameter risk stratification is the basis of counselling.

### Imaging

- **Multiparametric MRI (mpMRI)** = T2-weighted + diffusion-weighted + dynamic contrast-enhanced sequences → better locoregional characterisation.
- **Uses:** planning before definitive local therapy; **active surveillance candidacy**; **MRI–TRUS fusion targeted biopsy**.
- AUA and Society of Abdominal Radiology: **strongly consider high-quality MRI before repeat biopsy after a negative biopsy**; a **suspicious lesion warrants targeted biopsy**. The key points go further: **MRI before any biopsy**.
- **Staging is risk-based**, weighing pre-test probability of metastasis against test sensitivity and specificity:

| Asymptomatic low or intermediate risk                                  | High risk                      |
|------------------------------------------------------------------------|--------------------------------|
| Metastasis unlikely → <b>no routine CT abdomen–pelvis or bone scan</b> | <b>Bone scan + CT or mpMRI</b> |

- **Imaging is overused:** nodal disease is rare in contemporary screened populations.
- **Ultrasound advances** — colour and power Doppler, three-dimensional contrast Doppler, **HistoScanning** (computer-aided tissue characterisation) — remain investigational.
- **Prostate-specific membrane antigen (PSMA) positron emission tomography (PET):** <sup>68</sup>Ga-PSMA-11 and piflufolastat F-18 (<sup>18</sup>F-DCFPyL) are **FDA-approved for initial staging at high risk of metastasis**; **more sensitive and specific than conventional imaging**; benefit after negative conventional imaging unproven — consider in high risk.

**Concept** — *Staging imaging follows pre-test probability: yield is negligible in low- and intermediate-risk disease, so only high-risk disease is scanned.*

### Molecular Staging

#### COMPARISON — TISSUE-BASED GENOMIC TESTS

| Test     | Basis                                                    |
|----------|----------------------------------------------------------|
| Prolaris | <b>Cell-cycle progression</b> gene panel (proliferation) |

| Test                 | Basis                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------|
| Oncotype DX Prostate | Quantitative reverse-transcriptase polymerase chain reaction (RT-PCR): <b>17 genes, 5 pathways</b> |
| Decipher             | <b>22 RNA biomarkers</b> , multiple pathways                                                       |

- Variously predict **adverse pathology, biochemical recurrence, metastasis and cancer death**; may refine standard risk groups, but **need prospective validation**.

### Pelvic Lymphadenectomy

- **No consistent oncological benefit** from pelvic lymph node dissection (PLND); its value is **staging** — refined risk stratification, counselling and guiding selective secondary therapy.
- **Decide by nomogram-predicted risk** of nodal disease (guideline thresholds vary), balanced against PLND morbidity.

**Concept** — *PLND is a staging operation, not a therapeutic one; it is justified by the information it yields, so the decision rests on predicted nodal risk.*

#### KEY POINTS: DIAGNOSIS AND STAGING OF PROSTATE CANCER

- **Offer informed choice** on PSA testing; **combine PSA with DRE and clinical risk**.
- **Repeat PSA or add biomarkers** before biopsy; **MRI before biopsy** improves detection of significant cancer.
- **Assign a risk group at diagnosis** — it directs metastatic imaging and treatment.